

THANH LAM HO

HEAD OF ENGINEERING / TECHNICAL LEADER

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ABOUT ME

Technical Leader & System Architect with 14+ years designing and delivering large-scale web systems and, most recently, AI/RAG-powered products — from microservices architecture to production-grade DevOps pipelines and LLM-based platforms. Track record owning technical roadmaps, building and scaling engineering teams (up to 10 people), and leading high-traffic e-commerce/OMS and EdTech systems serving 25,000+ users at 10,000+ CCU. Hands-on experience architecting AI data pipelines, LLM gateways, and cost-aware model/vendor evaluation. Comfortable operating in ambiguous, fast-scaling environments, translating business goals and metrics into technical execution — and pushing back when needed. Seeking a Head of Engineering role to own technical direction end-to-end.

SKILLS

Backend: Java Spring Boot, PHP Laravel, Python Django

Architecture & DevOps: Microservices / System Design, Kubernetes / CI-CD, Docker / Linux

AI / Data: RAG architecture, LangGraph orchestration, Vector DB (Qdrant), LLM Gateway design, multi-tier model routing & cost optimization, Celery data pipelines

Database: MySQL / PostgreSQL, Redis, MongoDB

Frontend & API: ReactJS / HTML / CSS, REST API / GraphQL

Leadership: Hiring & Team Building, Mentoring & Performance Management, Agile / Scrum, Architecture Review, Budget & Vendor Evaluation (AI platform cost analysis)

WORK EXPERIENCES

Technical Leader

04/2023 - present

Digital Software of DAI TRUONG PHAT EDUCATION SJC (dtp-education.com)

Project: Chat Learning AI — RAG trợ lý học tập (AI initiative, 07/2026 - present)

Challenge: Design and build from scratch a grounded RAG assistant for K-6 curriculum content (Q&A, step-by-step problem solving, exam generation, AI-generated explainer videos), while keeping AI inference cost controllable and hallucination risk near zero.

- Architected end-to-end RAG pipeline: OCR ingestion → chunking → embedding → Qdrant vector store, orchestrated via LangGraph multi-node graph (retrieve → qa/solve/on_tap/exam_gen) with Redis-backed checkpointing
- Designed a provider-agnostic LLM Gateway (VNGCloud AI Platform / Anthropic & OpenAI protocols) with 2-tier model routing (cheap vs. strong) plus semantic caching — cut cost per repeated query by an estimated 80x+ on the expensive reasoning tier based on real token measurements
- Owned AI cost/vendor evaluation: measured real token usage per task, modeled monthly cost scenarios, and produced budget & quota recommendations before committing to the AI platform vendor
- Built deterministic safeguards against hallucination — citation-grounded answers, formula-verification guard for AI-generated videos, and code-based (not LLM-based) exam question-count validation against curriculum blueprints
- Set up automated quality evals (retrieval recall@5, deterministic matrix-match) run pre-merge to catch regressions when swapping models/prompts

Technology: FastAPI, LangGraph, Qdrant, PostgreSQL, Redis Stack, Celery, React/TypeScript, Docker, VNGCloud AI Platform (Gemini)

Project: APP EDUFUN (edufun.com.vn)

Challenge: Build a scalable English learning platform for K-12 students with rapidly growing user base.

- Architected microservices system now serving 25,000+ students at peak 10,000+ CCU
- Designed auto-scaling on Kubernetes ensuring high availability and zero-downtime deployments
- Built full CI/CD pipeline cutting release cycles from days to hours, supporting faster feature delivery during a high user-growth phase
- Recruited, onboarded and mentored a 6-member engineering team; set technical standards, ADRs, and led code reviews
- Worked with Product Owner to break down requirements into technical tasks and sprint plans, aligning engineering output with product/business priorities
- Team Size: 6 members | Position: Technical Leader

Technology: Spring Boot, Spring Security, Spring Data, Spring Cloud Gateway, Hibernate, Multi-threading | MySQL, Redis | Kubernetes | S3 | Restful API

Project: One Data — CMS for managing and publishing digital learning content

- Architected microservices and set up CI/CD pipelines on Kubernetes
- Led implementation and code reviews for a 4-member team
- Team Size: 4 members | Position: Technical Leader

Technology: Spring Boot (Java), Spring Security, Spring Data, Hibernate, Restful API, MySQL, Redis, S3

Project: LCMS (Learning Content Management System)

- Online English testing system handling up to 8,000 CCU
- Team Size: 6 members | Position: DevOps

Technology: Spring Boot (Java), Spring Security, Spring Data, Hibernate, Restful API, MySQL, Redis

Technical Leader

06/2022 - 04/2023

REE Digital — Second Tenure (reecorp.com)

- Returned to lead architecture reviews and system improvements on existing product suite
- Mentored junior engineers and established better CI/CD and deployment practices

Backend Leader

01/2020 - 06/2022

Pharmacy (pharmacy.vn)

Challenge: Scale a high-traffic e-commerce and OMS platform for the largest pharmacy chain in Vietnam.

Project: e-commerce (pharmacy.vn) + Order Management System (OMS)

- Led 3-member backend team — OMS processing 100,000+ orders/day, serving both retail storefront and B2B/partner order flows on a shared multi-tenant platform
- Scaled pharmacy.vn to handle 10,000+ CCU across 10,000+ SKUs with Elasticsearch integration, improving system stability during peak sales campaigns and reducing checkout-related incidents
- Resolved race conditions and enforced distributed-transaction consistency across the polyglot microservices stack during high-concurrency order and inventory updates
- Drove polyglot microservices migration: Laravel + Django + Spring Boot
- Defined team standards, conducted code reviews, mentored backend engineers
- Collaborated with cross-functional teams (Frontend, QA, DevOps, Product), translating business/operations requirements (fulfillment SLAs, promotion traffic spikes) into technical initiatives
- Team Size: 10 | Position: Backend Leader (Backend: 3 members)

Technology: Laravel (PHP), Django (Python), Spring Boot (Java), Restful API, GraphQL, MySQL, PostgreSQL, Kubernetes, S3, Elasticsearch

Freelance Backend Developer

09/2016 - 07/2017

Contract — Singapore client (Uplaucer project)

- Designed and developed backend features for Singapore-based client
- Managed full project lifecycle from requirements gathering to deployment

Technical Leader

06/2013 - 06/2017

REE Digital (reecorp.com)

Challenge: Build IoT monitoring and mobile apps from scratch for a fast-growing energy company.

Project: Reesolar System (monitoring.reesolar.com.vn)

- Hired, onboarded and led an 8-member engineering team from scratch to deliver a solar monitoring platform for 200+ IoT devices, 1,000+ users
- Reviewed architecture and code; defined coding standards for the team
- Team Size: 8 | Position: Technical Leader

Technology: Express (Node.js), Python, IoTCore, Google Cloud, ReactJS

Project: ReeTown APP

- Implemented features and led code review for 4-member mobile team
- Team Size: 4 | Position: Technical Leader

Technology: Spring Boot (Java), MySQL, Google Cloud, Flutter

Senior Web Developer

08/2017 - 08/2020

Lockon Viet Nam / YRGLM Vietnam

Part of the team developing AD Ebisu (ebis.ne.jp), Account Center, EC-CUBE e-commerce

- Developed marketing analytics platform "Ad Ebisu" — design docs, implementation, testing
- Led a sub-team of 2-3 engineers; reviewed code and wrote plugins for EC-CUBE
- Wrote unit tests with PHPUnit; integrated Restful API services

Technology: *Symfony 4, CakePHP, HTML5/CSS3, Bootstrap, jQuery, MySQL, PostgreSQL, Vertica, Restful API, PHPUnit, Git*

Senior Web Developer

08/2013 - 08/2017

SocialLOFT

Developing Moosocial (moosocial.com) and custom client projects

- Designed and implemented Restful API for mobile team; integrated Facebook, PayPal
- Implemented mobile app features (Xcode); deployed iOS apps to App Store
- Communicated with customers via JIRA and Skype

Technology: *CakePHP, HTML5/CSS3, Bootstrap, jQuery, RequireJS, MySQL, Google GEO API, Restful API, JIRA, Git*

Web Developer

05/2011 - 08/2013

YouNet

- Full stack developer on PHP social platforms (SocialEngine, phpFox)
- Customized social websites, integrated Google and Facebook plugins, set up servers
- Customized Zend Framework for client projects

Technology: *PHP, Zend Framework, Smarty, JavaScript, jQuery, HTML5/CSS3, Bootstrap, Facebook API, Google Map API, MySQL, SVN*

EDUCATION

Information Technology — Major: Information Systems

University of Science, Ho Chi Minh City 2006 - 2011

- Graduated with specialization in Information Systems
- Final-year project: Web-based social platform with PHP and MySQL
- Active member of university programming club

CERTIFICATIONS

Planned: AWS Solutions Architect / CKA (Kubernetes)

LANGUAGES

Vietnamese: Native · English: Professional Written Proficiency

SOFT SKILLS

Technical Mentoring, Architecture Decision, Sprint Planning, Stakeholder Communication, Code Review & Quality, Problem Solving, Cross-border collaboration (Singapore & Japan clients)

REFERENCES

Nguyễn Hữu Đạt — Founder & CEO, HanaCare (Former supervisor at Pharmacity / REE Digital)

0906.33.94.95 — Available upon request